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Availability-conditioned spatial compensation for traffic forecasting with incomplete observations

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Abstract

Accurate traffic forecasting depends on reliable historical observations, yet sensor readings are often unavailable in practice. Simply filling missing entries with numerical placeholders obscures their observation state and may distort the representation of historical traffic conditions. Graph neural networks can capture spatial dependencies among traffic sensors, but relying only on the physical road graph may not provide the most effective compensation when a sensor's own history is incomplete. We therefore propose the Availability-Conditioned Spatial Compensation Graph Convolutional Recurrent Network (ACSC-GCRN), which uses observation availability to guide both historical feature use and spatial compensation. An availability-gated encoder prevents unavailable values from contributing to the value representation while retaining the observation state. The model further complements the road graph with correlation and adaptive relations and adjusts their contributions according to the current observation condition. Experiments on METR-LA, PEMS04, and PEMS08 under standard and synthetic missingness protocols, together with an additional dataset-native invalid-reading protocol on METR-LA, show that ACSC-GCRN consistently remains among the two best-performing methods across diverse observation conditions. Ablation results support the gated encoding and multi-source spatial compensation, with availability-conditioned fusion showing greater value under severe missingness. These findings demonstrate that explicitly representing observation availability and using it to guide spatial compensation improves the robustness of traffic forecasting from incomplete histories.

Keywords Traffic forecasting, Incomplete observations, Spatio-temporal graph neural networks, Observation availability, Spatial compensation

Clinical trial number

Not applicable.

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1 Introduction

Traffic forecasting is a fundamental task in intelligent transportation systems, supporting applications such as traffic signal control, route planning, congestion management, and urban mobility analysis. The task aims to predict future traffic states, such as speed or flow, from historical observations collected by spatially distributed sensors. Since traffic data exhibit both temporal dynamics and spatial dependencies, effective forecasting models need to capture not only the temporal evolution of each sensor, but also the interactions among different locations in the traffic network.

Spatio-temporal graph neural networks have achieved strong performance in this task by representing traffic sensors as graph nodes and modeling their spatial interactions with graph-based operations. Representative methods include DCRNN, which models traffic diffusion with recurrent neural networks [1], STGCN, which combines graph convolution and temporal convolution [2], Graph WaveNet, which learns adaptive spatial dependencies [3], and AGCRN, which introduces node-adaptive graph convolutional recurrent modeling [4]. These methods have substantially improved traffic forecasting accuracy under standard experimental settings.

However, traffic forecasting models commonly assume that historical observations are available, whereas practical sensing systems may contain unavailable readings because of sensor faults, communication interruptions, or delayed transmission. Filling these entries with a fixed numerical placeholder does not resolve the resulting ambiguity: without an explicit observation-state indicator, the forecasting model cannot distinguish an unavailable reading from a genuine observation with the same value. Missingness should therefore be represented as an observation-unavailability state rather than treated as an ordinary numerical value.

Incomplete histories also change the spatial information required for forecasting. Graph neural networks can use information from related sensors, and the physical road graph provides a meaningful prior for this propagation. Nevertheless, when a sensor's own historical observations are incomplete, physical adjacency alone may not identify the nodes that provide the most informative compensation. Statistical similarities and latent dependencies can offer additional spatial evidence, and their usefulness may vary with the observation condition of each sample and node. Consequently, forecasting from incomplete histories involves two coupled questions: how observation availability should be represented, and how it should guide the selection of spatial compensation sources.

To address these questions, we propose the Availability-Conditioned Spatial Compensation Graph Convolutional Recurrent Network (ACSC-GCRN). At the feature level, an availability-gated encoder suppresses unavailable value contributions while preserving an explicit representation of the observation state. At the spatial level, the model complements the road graph with a training-set correlation graph and a learnable adaptive graph. A node-wise fusion mechanism adjusts the contributions of these spatial relations using historical observation availability and node embeddings. The fused spatial support is then incorporated into a graph convolutional recurrent backbone for direct multi-step forecasting without a separate history-imputation stage.

The main contributions of this work are summarized as follows:

- We explicitly represent observation availability for direct traffic forecasting from incomplete histories. The proposed gated encoding prevents unavailable placeholders

from entering the value representation while retaining the observation state as an input signal.

- We develop availability-conditioned, sample–node-specific spatial compensation over physical, statistical, and latent relations. This design allows different nodes and input samples to adjust their spatial information sources according to the current observation condition.
- We systematically evaluate ACSC-GCRN on three traffic datasets under the standard protocol, three synthetic missingness patterns, five missing rates, and the METR-LA dataset-native invalid-reading protocol. Comparisons with general and specialized missing-data baselines, together with controlled ablations, demonstrate stable performance across diverse observation conditions and verify the proposed mechanisms.

The remainder of this paper is organized as follows. Section 2 reviews spatio-temporal traffic forecasting and incomplete-data modeling. Section 3 formulates traffic forecasting from incomplete histories and defines the observation protocols. Section 4 introduces ACSC-GCRN. Section 5 presents the experimental settings and results. Section 6 discusses the findings and their scope, and Sect. 7 concludes the paper.

2 Related work

2.1 Spatio-temporal traffic forecasting

Traffic forecasting has been widely studied in intelligent transportation systems. Because traffic sensors are naturally distributed over road networks, graph-based spatio-temporal models have become an important class of forecasting methods. DCRNN models traffic flow as a diffusion process on directed graphs and combines diffusion convolution with recurrent neural networks [1]. STGCN uses graph convolution and temporal convolution to jointly capture spatial and temporal dependencies [2]. Later methods further improve the modeling of dynamic spatial dependencies. For example, ASTGCN introduces spatial and temporal attention mechanisms [5], Graph WaveNet learns an adaptive adjacency matrix together with dilated temporal convolution [3], and AGCRN uses learnable node embeddings to generate adaptive graphs and node-specific graph convolution parameters [4]. RST-GCNN further captures long-term spatio-temporal dependencies for parking availability prediction [6].

These methods have substantially improved traffic forecasting performance under standard settings. However, most of them are designed and evaluated under the assumption that the historical input sequence is fully observed. This assumption is often fragile in real-world traffic sensing systems, where sensor failures, communication interruptions, or delayed data transmission may lead to incomplete historical observations. When the input window is corrupted, directly applying a standard spatio-temporal forecasting model may be unreliable, because the model does not explicitly know which input values are real observations and which are missing placeholders.

Graph construction is another important issue in traffic forecasting. A predefined road graph provides useful physical connectivity, but it may not fully capture long-range correlations or latent functional similarities between sensors. Beyond static graph representations, temporal property graph models can further describe evolving graph structures [7]. Adaptive graph learning methods, such as Graph WaveNet and AGCRN, reduce the dependence on fixed topology by learning data-driven spatial dependencies [3, 4].

Nevertheless, relying on a single graph source may still be insufficient under missing historical inputs. Moreover, learned spatial supports are commonly shared across input samples and do not use the current observation condition to allocate different relational sources. In ACSC-GCRN, the road, correlation, and adaptive graphs serve as complementary candidate relations, while observation availability and node embeddings determine their sample–node-specific contributions.

The value of adaptive information fusion has also been demonstrated in other prediction and perception tasks. Lu et al. combine vibration and operating signals through interactive attention for remaining-useful-life prediction [8], while Zhang et al. integrate multi-scale features with fusion attention for lightweight weld defect detection [9]. Although their tasks and data modalities differ from traffic forecasting, these studies illustrate the usefulness of allocating heterogeneous evidence adaptively. Our work focuses on conditioning multiple spatial relations on observation availability in graph-based traffic forecasting.

2.2 Missing data modeling in time series and traffic data

Missing observations are common in real-world time series. Existing studies have proposed various methods to handle incomplete sequences. GRU-D incorporates missingness masks and time-decay mechanisms into recurrent neural networks [10]. BRITS learns missing values in a bidirectional recurrent framework [11]. More recent methods, such as SAITS and CSDI, use self-attention or diffusion-based generative modeling for time series imputation [12, 13]. In graph-structured time series, recent work has also started to address forecasting directly under missing inputs. For example, HD-TTS uses hierarchical spatio-temporal downsampling to forecast graph signals with missing observations [14], whereas BiTGraph incorporates missing patterns into biased temporal convolution and graph propagation [15]. We include both methods as specialized baselines for direct forecasting from incomplete histories.

Missing traffic data have also been studied with graph-based models. These methods usually exploit spatial correlations among sensors to estimate unavailable traffic observations. Graph Markov Network and GCN-based approaches use graph structures to model missing traffic data [16, 17], while memory-augmented dynamic graph models improve traffic data imputation by learning dynamic spatial relationships [18]. Recent studies further explore more complex missing-data scenarios. HSTGCN constructs hierarchical spatio-temporal graphs for traffic data imputation [19], and MagiNet introduces a mask-aware graph imputation network to learn representations from incomplete traffic data [20]. Other works combine imputation and prediction more directly. For instance, MTPF adopts multi-task pretraining and fine-tuning for traffic prediction and imputation [21], SEDA studies traffic prediction with missing data from a spatio-temporal ego-graph domain adaptation perspective [22], and ImFCP iteratively integrates traffic data completion and future prediction [23].

These studies are valuable for reconstructing incomplete traffic records or improving prediction with missing data. However, many of them still rely on explicit imputation, multi-task completion-prediction procedures, or additional adaptation mechanisms. Different from imputation-oriented studies, this paper focuses on forecasting future traffic states directly from incomplete historical inputs. The objective is not to recover a complete historical sequence before prediction, but to make the forecasting model

aware of missingness during prediction. Although direct forecasting methods such as HD-TTS and BiTGraph process incomplete graph sequences without a separate reconstruction stage, the joint use of observation availability to control historical feature use and allocate multiple spatial relations remains underexplored. ACSC-GCRN addresses this problem by combining availability-gated input encoding with availability-conditioned, node-wise spatial compensation.

3 Problem formulation

This section formulates traffic forecasting with incomplete historical observations. We first introduce the standard traffic forecasting task and then define incomplete historical inputs together with their observation-availability masks. In addition, we define three controlled synthetic missingness patterns to simulate common missing-data scenarios in real-world situations.

3.1 Traffic forecasting

Given a road network consisting of N traffic sensor nodes, we represent it as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, where $\mathcal{V}$ denotes the set of sensor nodes with $|\mathcal{V}| = N$, $\mathcal{E}$ denotes the set of edges, and $\mathbf{A} \in \mathbb{R}_{\geq 0}^{N \times N}$ denotes a general weighted adjacency matrix describing the spatial dependencies among the nodes. The matrix $\mathbf{A}$ is not restricted to binary physical connectivity; each element a_{ij} represents the dependency strength from node j to node i. In the proposed model, the spatial dependency matrix is instantiated for each input sample through an availability-conditioned, node-wise fusion of a predefined road graph, a correlation graph, and an adaptive graph, as detailed in Sect. 4. The traffic observations at time step t are denoted by

$$\mathbf{X}_t \in \mathbb{R}^{N \times C}, \tag{1}$$

where C denotes the number of features associated with each node. These features may include traffic speed, traffic flow, road occupancy, or other traffic-related variables.

Given a historical observation sequence of length P:

$$\mathbf{X}_{t-P+1:t} = [\mathbf{X}_{t-P+1}, \mathbf{X}_{t-P+2}, \dots, \mathbf{X}_t] \in \mathbb{R}^{P \times N \times C}, \tag{2}$$

the objective of traffic forecasting is to predict the traffic states over the next Q time steps:

$$\mathbf{Y}_{t+1:t+Q} = [\mathbf{X}_{t+1}, \mathbf{X}_{t+2}, \dots, \mathbf{X}_{t+Q}] \in \mathbb{R}^{Q \times N \times C}. \tag{3}$$

Under the assumption of complete historical observations, the traffic forecasting model can be formulated as:

$$\hat{\mathbf{Y}}_{t+1:t+Q} = f_\Theta(\mathbf{X}_{t-P+1:t}, \mathcal{A}), \tag{4}$$

where $f_\Theta(\cdot)$ denotes the forecasting model parameterized by Θ, and $\hat{\mathbf{Y}}_{t+1:t+Q}$ denotes the predicted future traffic states produced by the model.

3.2 Traffic forecasting with incomplete historical observations

Standard traffic forecasting commonly assumes that every value in the historical input $\mathbf{X}_{t-P+1:t}$ is available. In practical traffic sensing systems, however, observations may be

unavailable because of sensor failures, communication interruptions, or delayed transmission. An unavailable entry is not an ordinary numerical value but an observation state indicating that the corresponding traffic value cannot be used as measured evidence. To explicitly represent this state, we introduce a binary observation-availability mask:

$$\mathbf{M}_{t-P+1:t} \in \{0,1\}^{P\times N\times C}, \tag{5}$$

where its element $m_{\tau,i,c}$ is defined as

$$m_{\tau,i,c} = \begin{cases} 1, & \text{if } x_{\tau,i,c} \text{ is available,} \\ 0, & \text{if } x_{\tau,i,c} \text{ is unavailable.} \end{cases} \tag{6}$$

Here, $\tau \in \{t-P+1,\dots,t\}$ denotes a time step within the historical input window, $i \in \{1,\dots,N\}$ denotes the index of a traffic sensor node, and $c \in \{1,\dots,C\}$ denotes the index of a traffic feature. The scalar $x_{\tau,i,c}$ denotes the c-th traffic feature observed at node i and time step τ, and is an element of the historical observation tensor $\mathbf{X}_{t-P+1:t}$. Accordingly, $m_{\tau,i,c}$ explicitly indicates whether the corresponding historical value is available to the forecasting model.

According to the missingness mask, the incomplete historical input is represented as

$$\widetilde{\mathbf{X}}_{t-P+1:t} = \mathbf{X}_{t-P+1:t} \odot \mathbf{M}_{t-P+1:t}, \tag{7}$$

where $\odot$ denotes element-wise multiplication. Zeros are used only as numerical placeholders for unavailable entries. Because a placeholder zero may also be a valid observed value, the model receives the availability mask $\mathbf{M}_{t-P+1:t}$ together with $\widetilde{\mathbf{X}}_{t-P+1:t}$ so that numerical values can be distinguished from their observation states.

Therefore, the traffic forecasting task under missing historical observations is formulated as

$$\widehat{\mathbf{Y}}_{t+1:t+Q} = f_{\Theta}\left(\widetilde{\mathbf{X}}_{t-P+1:t}, \mathbf{M}_{t-P+1:t}, \mathcal{A}\right). \tag{8}$$

It should be emphasized that this study focuses on future traffic forecasting under missing historical inputs rather than the imputation of missing historical values. Therefore, missingness is introduced only into the historical input window $\mathbf{X}_{t-P+1:t}$, while the future prediction target $\mathbf{Y}_{t+1:t+Q}$ remains unchanged during training and evaluation. This setting is consistent with practical deployment scenarios, in which the model is required to predict future network-wide traffic states from incomplete historical sensor observations.

3.3 Synthetic missing patterns

To systematically evaluate forecasting robustness under controlled incomplete inputs, we consider three synthetic missingness patterns: random missingness, block missingness, and node-level missingness. These three patterns correspond to different forms of data anomalies found in real-world transportation systems. All patterns are applied only to the historical input window, while the future prediction targets are not artificially corrupted.

Random Missingness: Random missingness refers to the scenario in which observations at a subset of time–node positions within the historical input window are randomly

unavailable. Given a missing rate r, the random missingness pattern can be formulated as:

$$\mathbf{m}_{\tau,i,c} \sim \text{Bernoulli}(1-r), \tag{9}$$

where $r \in [0,1]$ denotes the target proportion of missing observations. This setting is used to simulate random packet loss, transient communication failures, or sporadic data acquisition failures in traffic sensing systems.

Block Missingness: Continuous block missingness refers to the scenario in which one or more consecutive temporal segments in the historical input are unavailable. Compared with random missingness, continuous block missingness disrupts the local temporal continuity of traffic sequences. Formally, given a starting time τ_s and a block length L_b, continuous block missingness can be represented as:

$$\mathbf{m}_{\tau,i,c} = 0, \quad \tau \in [\tau_s, \tau_s + L_b - 1], \tag{10}$$

where L_b denotes the duration of the consecutive missing interval. This setting is used to simulate communication interruptions, batch data-upload failures, or temporary device malfunctions.

Node Missingness: Node-level missingness refers to the scenario in which a subset of sensor nodes is unavailable throughout the entire historical input window. Given a set of missing nodes $\mathcal{V}_m \subseteq \mathcal{V}$, node-level missingness can be represented as:

$$\mathbf{m}_{\tau,i,c} = 0, \quad i \in \mathcal{V}_m,\ \tau \in [t-P+1, t]. \tag{11}$$

This setting is used to simulate sensor failures, device maintenance, or prolonged observation interruptions in local regions. Because the historical information of the target nodes may be entirely unavailable, node-level missingness places greater demands on the model's ability to recover information from spatial dependencies.

The three missingness patterns described above characterize different forms of incomplete historical observations. Random missingness primarily evaluates the model's robustness to locally scattered missing observations; continuous block missingness assesses its forecasting capability when temporal continuity is disrupted; and node-level missingness examines whether the model can compensate for unavailable observations by exploiting information from other spatially correlated nodes. In the subsequent experiments, the effectiveness of the proposed method is systematically evaluated under these three missing-data scenarios.

4 Methodology

This section presents the Availability-Conditioned Spatial Compensation Graph Convolutional Recurrent Network (ACSC-GCRN). Building on an AGCRN-style forecasting backbone [4], ACSC-GCRN uses observation availability to condition both historical representation and spatial information aggregation. It employs a gated mechanism to jointly encode the filled observations and their availability masks, while integrating predefined, correlation, and adaptive spatial relations through sample–node-specific graph-source weights. The resulting availability-aware representation and fused spatial support are processed by the recurrent forecasting backbone to generate multi-step predictions directly, without a separate history-imputation stage. Figure 1 illustrates the overall framework.

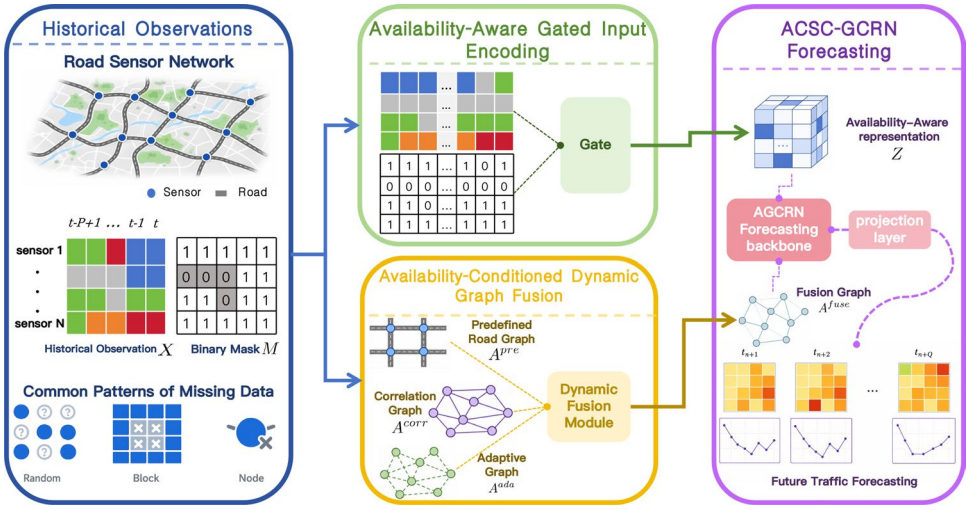


Fig. 1 Overview of ACSC-GCRN. Observation availability controls both the historical representation and the node-wise allocation of spatial supports; no separate imputation target is introduced

4.1 Availability-aware gated input encoding

In scenarios with missing historical observations, if missing entries are simply set to zero and then fed into the forecasting model, the model cannot determine whether a zero value in the input corresponds to a genuine observation or a placeholder for a missing value. This semantic ambiguity may affect the model's understanding of historical traffic states. To address this issue, we treat the observation-availability mask as an additional input signal and jointly encode it with the historical observations after missing values have been filled.

For an arbitrary historical time step $\tau \in [t-P+1, t]$, the observation after missing-value filling is denoted as $\widetilde{\mathbf{X}}_\tau \in \mathbb{R}^{N\times C}$, and the corresponding observation-availability mask is denoted as $\mathbf{M}_\tau \in \{0,1\}^{N\times C}$. We apply separate linear mappings to the observations and the observation-availability mask:

$$\mathbf{E}_\tau^x = \widetilde{\mathbf{X}}_\tau \mathbf{W}_x + \mathbf{b}_x, \tag{12}$$

$$\mathbf{E}_\tau^m = \mathbf{M}_\tau \mathbf{W}_m + \mathbf{b}_m, \tag{13}$$

where $\mathbf{W}_x \in \mathbb{R}^{C\times d}$ and $\mathbf{W}_m \in \mathbb{R}^{C\times d}$ are learnable parameters, and d denotes the hidden dimension. Their interaction is then modeled by a feature-wise gate:

$$\mathbf{G}_\tau = \sigma\left([\mathbf{E}_\tau^x \| \mathbf{E}_\tau^m]\mathbf{W}_g + \mathbf{b}_g\right), \tag{14}$$

$$\mathbf{Z}_\tau = \mathbf{M}_\tau \odot \mathbf{G}_\tau \odot \mathbf{E}_\tau^x + \mathbf{E}_\tau^m, \tag{15}$$

where $\mathbf{W}_g \in \mathbb{R}^{2d\times d}$ and $\mathbf{b}_g \in \mathbb{R}^d$ are learnable parameters. Here, $\sigma(\cdot)$ denotes the sigmoid function and $\|$ denotes feature concatenation. For the scalar traffic variable considered in this study, each availability indicator is applied to all d dimensions of the corresponding node representation. The availability mask determines whether the value representation contributes to the encoded input, while the learned gate controls the contribution of available observations. Consequently, the value branch is suppressed at unavailable positions, whereas the availability representation is retained in both states.

Accordingly, the input representation of the complete historical window is given by:

$$\mathbf{Z}_{t-P+1:t} = [\mathbf{Z}_{t-P+1}, \mathbf{Z}_{t-P+2}, \dots, \mathbf{Z}_t] \in \mathbb{R}^{P\times N\times d}. \tag{16}$$

This design enables the model to access not only the numerical input after missing-value filling but also the explicit availability of observations at each time–node position. Unlike imputation-then-prediction methods, ACSC-GCRN does not require reconstruction of a complete historical sequence at the input stage. Instead, it directly incorporates observation availability into the forecasting model, allowing the subsequent spatio-temporal encoding process to adjust information use based on observation availability.

4.2 Multi-source spatial graph construction

Spatial dependencies relevant to traffic forecasting may arise from physical connectivity, statistical similarity, and latent functional relations. Accordingly, we construct three candidate spatial graphs: a predefined road graph $\mathbf{A}^{pre}$, a correlation graph $\mathbf{A}^{corr}$, and an adaptive graph $\mathbf{A}^{ada}$. The road graph provides a fixed physical prior, while the correlation and adaptive graphs complement it with data-driven relations that can support spatial compensation when local historical observations are incomplete. These candidate graphs are subsequently integrated according to observation availability.

4.2.1 Predefined road graph

The predefined road graph $\mathbf{A}^{pre}$ is constructed from the road topology, sensor distances, or adjacency relationships provided by the dataset. This graph reflects the physical spatial relationships among traffic sensors and provides a stable road-network prior for the model. Given the original adjacency matrix, we set its diagonal entries to one to ensure self-loops and then apply row normalization:

$$\bar{\mathbf{A}}^{pre} = \mathrm{RowNorm}\left(\mathrm{SetDiag}_1(\mathbf{A}^{pre})\right), \tag{17}$$

where $\mathrm{SetDiag}_1$ denotes setting all diagonal elements to 1, and $\mathrm{RowNorm}(\cdot)$ denotes the row-normalization operation. The predefined road graph is mainly used to characterize local traffic propagation relationships between physically adjacent nodes.

4.2.2 Correlation graph

Relying solely on road topology may be insufficient to capture statistical similarities between nodes that are not directly adjacent. Therefore, we further construct a node-correlation graph based on historical traffic sequences in the training set. Let the historical sequences of node i and node j in the training set be denoted as $\mathbf{s}_i = [s_{i,1}, s_{i,2}, \dots, s_{i,L}]$ and $\mathbf{s}_j = [s_{j,1}, s_{j,2}, \dots, s_{j,L}]$, respectively, where L denotes the number of training time steps used for correlation calculation, and $s_{i,l}$ denotes the observation of node i at the -th training time step. Their Pearson correlation coefficient is defined as:

$$\rho_{ij} = \frac{\sum_l (s_{i,l} - \bar{s}_i)(s_{j,l} - \bar{s}_j)}{\sqrt{\sum_l (s_{i,l} - \bar{s}_i)^2}\sqrt{\sum_l (s_{j,l} - \bar{s}_j)^2}}, \tag{18}$$

where $\bar{s}_i$ and $\bar{s}_j$ denote the mean values of the historical sequences of node i and node j, respectively.

To avoid introducing excessive weakly correlated edges, we retain only the top-k most correlated neighbors for each node. The node-correlation graph is defined as:

$$\mathbf{A}_{ij}^{corr} = \begin{cases} |\rho_{ij}|, & j \in \mathrm{TopK}(i), \\ 0, & \text{otherwise}, \end{cases} \tag{19}$$

where $\mathrm{TopK}(i)$ denotes the set of the k nodes with the largest absolute correlations to node i. We then add self-loops and apply row normalization in the same manner:

$$\bar{\mathbf{A}}^{corr} = \mathrm{RowNorm}\left(\mathbf{A}^{corr} + \mathbf{I}\right). \tag{20}$$

It should be emphasized that the node-correlation graph is constructed only from the training set; the validation and test sets are not involved in correlation estimation, thereby avoiding information leakage. This graph complements the road topology by capturing long-range correlations and functional similarities that cannot be explicitly represented by physical adjacency alone.

4.2.3 Adaptive graph

In addition to the predefined road graph and the correlation graph, we introduce an adaptive graph to represent latent spatial dependencies. Let the node embedding matrix be denoted as

$$\mathbf{E} \in \mathbb{R}^{N \times d_e}, \tag{21}$$

where d_e denotes the embedding dimension. The embedding matrix is randomly initialized and jointly optimized with the forecasting network on the training set. It is therefore not a fixed embedding extracted from the road topology or separately computed traffic statistics.

$$\mathbf{A}^{ada} = \mathrm{Softmax}\left(\mathrm{ReLU}\left(\mathbf{E}\mathbf{E}^{\top}\right)\right), \tag{22}$$

where $\mathrm{Softmax}(\cdot)$ is computed along the row dimension so that the dependency weights from each node to all other nodes sum to 1. This design follows the idea of adaptive graph learning [3, 4], enabling the model to automatically learn latent spatial relationships beyond those captured by the predefined graph and the correlation graph during training.

4.3 Availability-conditioned dynamic graph fusion

The three graph structures characterize different types of spatial dependencies. The predefined road graph has stable physical semantics, the node-correlation graph reflects statistical similarities, and the adaptive graph learns latent data-driven relationships. However, their relative contributions may vary across input samples and sensor nodes when the availability of local historical observations changes. Therefore, instead of sharing one set of graph weights across the entire dataset, we determine the contribution of each graph source according to the observation availability of each sample and node.

Given the set of candidate graphs:

$$\mathcal{A} = \left\{\bar{\mathbf{A}}^{pre}, \bar{\mathbf{A}}^{corr}, \mathbf{A}^{ada}\right\}, \tag{23}$$

we index the three candidate graphs as $\mathbf{A}^{(1)} = \bar{\mathbf{A}}^{pre}$, $\mathbf{A}^{(2)} = \bar{\mathbf{A}}^{corr}$, and $\mathbf{A}^{(3)} = \mathbf{A}^{ada}$.

For input sample b, the historical observation ratio of node i is defined as

$$r_{b,i} = \frac{1}{PC} \sum_{\tau=t-P+1}^{t} \sum_{c=1}^{C} m_{\tau,i,c}^{(b)}, \tag{24}$$

where b indexes the input samples and $m_{\tau,i,c}^{(b)}$ is the availability indicator of feature c at node i and historical time step τ in sample b. Here, P is the length of the historical input window and C is the number of traffic features, so PC is the total number of possible observations of a node within the window. The resulting ratio $r_{b,i} \in [0, 1]$ represents the proportion of available historical observations of node i in sample b.

The observation ratio is combined with the node embedding to generate sample–node-specific graph weights:

$$\alpha_{b,i} = \mathrm{Softmax}\left(\mathrm{MLP}_{\phi}\left([\mathbf{e}_i \| r_{b,i}]\right)\right), \tag{25}$$

where $\mathbf{e}_i \in \mathbb{R}^{d_e}$ is the i-th row of the node embedding matrix $\mathbf{E}$, $\|$ denotes feature concatenation, and $\mathrm{MLP}_{\phi} : \mathbb{R}^{d_e+1} \to \mathbb{R}^3$ is a learnable mapping with parameters ϕ, shared across all samples and nodes. The softmax operation is applied over the three graph sources. Thus, $\alpha_{b,i} \in \mathbb{R}^3$ contains the graph-source weights for node i in sample b, and $\alpha_{b,i,g}$ denotes its g-th element. The weights satisfy $\alpha_{b,i,g} \geq 0$ and $\sum_{g=1}^{3} \alpha_{b,i,g} = 1$.

The fused spatial dependency matrix for sample b is defined as

$$\left[\mathbf{A}^{fuse,(b)}\right]_{ij} = \sum_{g=1}^{3} \alpha_{b,i,g} \left[\mathbf{A}^{(g)}\right]_{ij}. \tag{26}$$

Here, i and j index the sensor nodes, $g \in \{1, 2, 3\}$ indexes the predefined, correlation, and adaptive graphs, respectively, and $[\mathbf{A}^{(g)}]_{ij}$ denotes the (i, j)-th element of the corresponding adjacency matrix.

Since the graph-source weights are nonnegative and sum to one, each row of the fused graph is a convex combination of the corresponding rows of the three candidate graphs. The resulting matrix $\mathbf{A}^{fuse,(b)} \in \mathbb{R}^{N \times N}$ varies across input samples, and different nodes within the same sample may assign different weights to the three spatial sources. Because $r_{b,i}$ summarizes observation availability over the complete historical window, the fusion is dynamic at the input-window level rather than at each individual time step. The same fused graph is used across the recurrent updates for a given input window.

4.4 Spatiotemporal forecasting

After the availability-aware input encoding and dynamic graph fusion, the model obtains two inputs for each sample b: the historical representation $\mathbf{Z}_{t-P+1:t}^{(b)} \in \mathbb{R}^{P \times N \times d}$ and the fused spatial dependency matrix $\mathbf{A}^{fuse,(b)} \in \mathbb{R}^{N \times N}$. The former represents the historical traffic values together with their observation availability, while the latter integrates the predefined, correlation, and adaptive spatial relations through sample–node-specific weights conditioned on observation availability and node embeddings. ACSC-GCRN feeds them into an AGCRN-style graph convolutional recurrent backbone to model spatiotemporal dependencies.

The graph convolution operation aggregates information among related traffic nodes according to $\mathbf{A}^{fuse,(b)}$, while the recurrent updates capture temporal dependencies over

the historical window. Unlike standard AGCRN, the forecasting backbone operates on the availability-aware representation rather than the original historical observations and uses the conditionally fused graph as its spatial support.

Figure 2 illustrates the spatiotemporal forecasting backbone. The encoded historical sequence is processed sequentially within each recurrent layer and then passed layer by layer through the encoder. Within each cell, the fused graph and node embeddings condition the gate and candidate-state computations, and the final hidden state of the top layer is projected directly to the Q-step forecast.

For notational simplicity, the layer index is omitted in the following recurrent-update equations. The equations describe the update of the first recurrent layer, which takes $\mathbf{Z}_{\tau}^{(b)}$ as input. Higher recurrent layers follow the same formulation but take the hidden states produced by the preceding layer as their inputs. For sample b, the first recurrent layer updates its hidden state at each historical time step τ as

$$\mathbf{H}_{\tau}^{(b)} = \mathrm{AGCRNCell}\left(\mathbf{Z}_{\tau}^{(b)}, \mathbf{H}_{\tau-1}^{(b)}; \mathbf{A}^{fuse,(b)}, \mathbf{E}\right), \tag{27}$$

where $\mathbf{H}_{\tau}^{(b)} \in \mathbb{R}^{N \times d_h}$ denotes the hidden state of the first recurrent layer for sample b at time step τ, d_h is the hidden-state dimension, $\mathbf{E} \in \mathbb{R}^{N \times d_e}$ is the learnable node embedding matrix, and AGCRNCell$(\cdot)$ denotes the graph convolutional recurrent cell.

Specifically, the graph convolution operation inside the recurrent cell is implemented by an adaptive vertex-wise graph convolution network AVWGCN$(\cdot)$. AVWGCN is the basic graph convolution operation used in AGCRN. It uses $\mathbf{A}^{fuse,(b)}$ as the spatial support and generates node-specific transformation parameters from the node embedding matrix $\mathbf{E}$. This allows different traffic sensors to learn different local mapping functions while aggregating information through the conditionally fused spatial dependencies.

The gated recurrent update is formulated as

$$[\mathbf{R}_{\tau}^{(b)}, \mathbf{U}_{\tau}^{(b)}] = \sigma\left(\mathrm{AVWGCN}\left([\mathbf{Z}_{\tau}^{(b)}, \mathbf{H}_{\tau-1}^{(b)}]; \mathbf{A}^{fuse,(b)}, \mathbf{E}\right)\right), \tag{28}$$

$$\widetilde{\mathbf{H}}_{\tau}^{(b)} = \tanh\left(\mathrm{AVWGCN}\left([\mathbf{Z}_{\tau}^{(b)}, \mathbf{R}_{\tau}^{(b)} \odot \mathbf{H}_{\tau-1}^{(b)}]; \mathbf{A}^{fuse,(b)}, \mathbf{E}\right)\right), \tag{29}$$

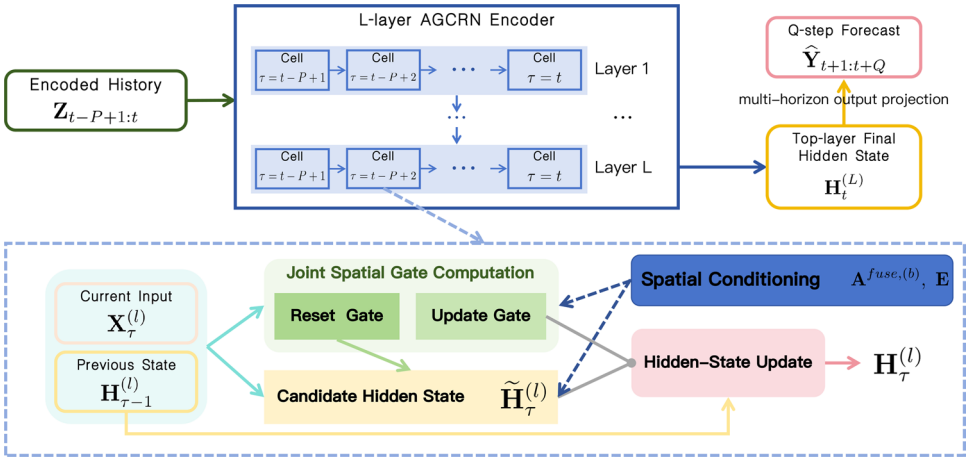


Fig. 2 Spatiotemporal forecasting backbone of ACSC-GCRN. The upper part shows the stacked recurrent encoder and direct multi-horizon projection, while the lower part summarizes the main information flow within an AGCRN cell

$$\mathbf{H}_{\tau}^{(b)} = \mathbf{U}_{\tau}^{(b)} \odot \mathbf{H}_{\tau-1}^{(b)} + \left(1 - \mathbf{U}_{\tau}^{(b)}\right) \odot \widetilde{\mathbf{H}}_{\tau}^{(b)}, \tag{30}$$

where $\mathbf{R}_{\tau}^{(b)}$ and $\mathbf{U}_{\tau}^{(b)}$ denote the reset and update gates, respectively, and $\widetilde{\mathbf{H}}_{\tau}^{(b)}$ denotes the candidate hidden state. The functions $\sigma(\cdot)$ and $\tanh(\cdot)$ denote the sigmoid and hyperbolic tangent activations, $\odot$ denotes element-wise multiplication, and $[\cdot,\cdot]$ denotes feature concatenation. The AVWGCN operators used for the gate and candidate-state computations have separate learnable parameters, although both use $\mathbf{A}^{fuse,(b)}$ as the spatial support and $\mathbf{E}$ for node-specific parameter generation.

Let L denote the number of recurrent layers. After the complete historical window has been processed through all recurrent layers, the final hidden state of the top layer, $\mathbf{H}_{t}^{(b,L)}$, is used to generate the future traffic prediction:

$$\widehat{\mathbf{Y}}_{t+1:t+Q}^{(b)} = \mathrm{Proj}\left(\mathbf{H}_{t}^{(b,L)}\right), \tag{31}$$

where $\mathrm{Proj}(\cdot)$ denotes the output projection layer. It maps the final hidden representation to $\mathbb{R}^{Q\times N\times C}$, corresponding to the predicted traffic states of N nodes over the next Q time steps.

Through this forecasting backbone, ACSC-GCRN integrates the availability-aware historical representation, the sample-specific fused spatial dependency matrix, and the node-adaptive transformations generated from $\mathbf{E}$. The model therefore produces future traffic predictions directly from incomplete historical observations without reconstructing the unavailable input values.

4.5 Training objective

The model is trained in a supervised learning manner. Given the training sample set $\mathcal{D}_{train}$, each sample consists of the historical input after missing-value filling, the observation-availability mask, and the future ground-truth labels:

$$\left(\widetilde{\mathbf{X}}_{t-P+1:t}, \mathbf{M}_{t-P+1:t}, \mathbf{Y}_{t+1:t+Q}\right). \tag{32}$$

We use the mean absolute error as the primary training objective. Given the prediction $\widehat{\mathbf{Y}}$ and the ground truth $\mathbf{Y}$, the loss function is defined as:

$$\mathcal{L} = \frac{1}{QNC} \sum_{q=1}^{Q} \sum_{i=1}^{N} \sum_{c=1}^{C} \left|\widehat{\mathbf{Y}}_{q,i,c} - \mathbf{Y}_{q,i,c}\right|. \tag{33}$$

Controlled synthetic missingness is introduced only into the historical input and is represented by the observation-availability mask $\mathbf{M}$. The synthetic mask is generated according to the prescribed random, block, or node-level pattern, independently of the numerical input values, and no synthetic masking is applied to the future target.

Separately, to exclude target positions designated as invalid under the training and evaluation configuration, we use a target-validity mask $\mathbf{V} \in \{0,1\}^{Q\times N\times C}$. It is determined from the raw-scale future target before normalization as

$$V_{q,i,c} = \begin{cases} 1, & Y_{q,i,c}^{\mathrm{raw}} \neq v_{\mathrm{null}}, \\ 0, & Y_{q,i,c}^{\mathrm{raw}} = v_{\mathrm{null}}. \end{cases} \tag{34}$$

where v_{null} denotes the null value specified in the dataset metadata. In the training and evaluation configurations used in this study, the metadata for all three datasets specify $v_{\text{null}} = 0.0$. This convention is used only to determine valid target positions for loss and metric computation; it does not define the synthetic missingness of the historical input. The mask $\mathbf{V}$ is neither learned nor provided as an input to the forecasting model. The resulting masked MAE is

$$\mathcal{L}_{mask} = \frac{\sum_{q=1}^{Q}\sum_{i=1}^{N}\sum_{c=1}^{C} V_{q,i,c}\left|\widehat{\mathbf{Y}}_{q,i,c} - \mathbf{Y}_{q,i,c}\right|}{\sum_{q=1}^{Q}\sum_{i=1}^{N}\sum_{c=1}^{C} V_{q,i,c}}. \tag{35}$$

When all future target positions are valid, $\mathbf{V}$ is an all-one matrix and $\mathcal{L}_{mask}$ reduces to the standard mean absolute error.

The overall training objective is formulated as:

$$\Theta^{*} = \arg\min_{\Theta} \frac{1}{|\mathcal{D}_{train}|} \sum_{(\widetilde{\mathbf{X}},\mathbf{M},\mathbf{Y})\in\mathcal{D}_{train}} \mathcal{L}_{mask}\left(f_{\Theta}(\widetilde{\mathbf{X}},\mathbf{M},\mathcal{A}),\mathbf{Y};\mathbf{V}\right). \tag{36}$$

The historical observation-availability mask $\mathbf{M}$ and the target-validity mask $\mathbf{V}$ serve different purposes. $\mathbf{M}$ represents the availability of historical observations and participates in the availability-aware input encoding and dynamic graph fusion, whereas $\mathbf{V}$ is not provided to the forecasting model and is used only to exclude invalid future targets from the training loss and evaluation metrics. The training loss is optimized on normalized data, whereas the evaluation metrics are computed on the original traffic scale after inverse normalization.

5 Experiments

This section evaluates the effectiveness of ACSC-GCRN on traffic forecasting tasks through experiments. The experiments are designed to answer the following research questions:

(1) **RQ1:** Can ACSC-GCRN maintain competitive forecasting performance under the standard protocol without additional synthetic masking?
(2) **RQ2:** How robust is ACSC-GCRN to different synthetic missingness patterns and rates compared with both general forecasting models and methods specifically designed for incomplete traffic data?
(3) **RQ3:** Does the effectiveness of ACSC-GCRN extend to dataset-native invalid readings when observation availability is determined from the dataset rather than an artificially generated mask?
(4) **RQ4:** How do the availability-aware input encoding, data-driven spatial graph sources, and availability-conditioned dynamic graph fusion contribute to forecasting performance, and do their benefits become more evident under severe missingness?

5.1 Datasets

We evaluate the proposed method on three real-world traffic forecasting datasets, including METR-LA, PEMS04, and PEMS08. METR-LA contains traffic speed measurements collected from 207 loop detectors in Los Angeles County, while PEMS04 and PEMS08 contain traffic flow measurements collected from the Caltrans Performance

Measurement System. All datasets are sampled every 5 minutes and are provided with predefined graph structures that describe the spatial relationships among sensors.

Following common traffic forecasting settings, we use the previous 12 time steps to predict the next 12 time steps. Since the sampling interval is 5 minutes, both the input and prediction horizons correspond to one hour. The datasets are chronologically split into training, validation, and test sets according to the ratios shown in Table 1.

Controlled synthetic missingness is introduced after data splitting and is applied only to the historical input window. The future targets are not synthetically modified, while dataset-defined invalid target positions are excluded using the target-validity mask $\mathbf{V}$ defined in Sect. 4.5. For a given dataset, sample, missingness pattern, missing rate, and random seed, all compared methods use the same deterministic missing-mask realization. Normalization statistics and the correlation graph are estimated exclusively from the training split to prevent information leakage.

5.2 Baselines

We compare ACSC-GCRN with ten representative forecasting baselines, including eight general forecasting models and two methods specifically designed for incomplete traffic data. The general baselines cover graph-based, linear, MLP-based, spectral, and Transformer-based forecasting paradigms. AGCRN is a graph-based traffic forecasting model that learns adaptive node embeddings and node-specific graph convolution parameters to capture spatial heterogeneity without relying on a predefined graph [4]. DLinear and NLinear are simple yet strong linear forecasting baselines from LTSF-Linear. DLinear decomposes the input sequence into trend and seasonal components and applies linear mappings for forecasting, while NLinear uses a normalization strategy based on the last observed value to alleviate distribution shift [24]. STID is a lightweight MLP-based spatio-temporal forecasting model that uses identity embeddings to model spatial and temporal patterns. In our implementation, we use its spatial identity component under the unified input protocol [25]. StemGNN models multivariate time series by jointly capturing inter-series correlations and temporal dependencies in the spectral domain through graph Fourier transform and discrete Fourier transform [26]. TimesNet transforms one-dimensional time series into two-dimensional temporal representations according to multiple periods and uses TimesBlock to model complex temporal variations [27]. PatchTST is a Transformer-based model that segments time series into patches and adopts a channel-independent design to improve long-term forecasting [28]. iTransformer further revisits the Transformer architecture by treating variables as tokens and applying attention along the variate dimension to capture multivariate correlations [29].

In addition to these general forecasting models, we include BiTGraph and HD-TTS as baselines specifically designed for incomplete traffic data. BiTGraph explicitly incorporates missing patterns into biased temporal and graph convolutions [15], whereas HD-TTS uses hierarchical spatio-temporal representations to forecast graph signals with missing observations [14].

Table 1 Statistics of the datasets used in the experiments

Dataset	Variable	Nodes	Time steps	Split ratio	Sampling interval
METR-LA	Speed	207	34,272	7:1:2	5 min
PEMS04	Flow	307	16,992	6:2:2	5 min
PEMS08	Flow	170	17,856	6:2:2	5 min

The implementations of all methods are adapted to the common data and evaluation protocol used in this study, including the same chronological data splits, forecasting setting, historical missingness conditions, and metric definitions. Together, these baselines provide comparisons across different forecasting paradigms under the standard protocol without additional synthetic masking, controlled synthetic missingness, and the METR-LA dataset-native invalid-reading protocol.

5.3 Evaluation metrics

We evaluate forecasting performance using three commonly used metrics: mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE). Lower values indicate better forecasting accuracy. All metrics are computed on the inverse-normalized predictions, so the reported results are measured on the original traffic scale.

Let $\hat{\mathbf{Y}}$ and $\mathbf{Y}$ denote the predicted and ground-truth future traffic series, respectively. Consistent with the target-validity mask $\mathbf{V}$ defined in Sect. 4.5, we calculate the evaluation metrics only at valid target positions. Let $\Omega = \{i \mid V_i = 1\}$ denote the set of valid positions, where i is a flattened index over all prediction-time, node, and feature dimensions. The metrics are computed as follows:

$$\mathrm{MAE} = \frac{1}{|\Omega|} \sum_{i \in \Omega} \left| \hat{Y}_i - Y_i \right|, \tag{37}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{|\Omega|} \sum_{i \in \Omega} \left(\hat{Y}_i - Y_i \right)^2}, \tag{38}$$

$$\mathrm{MAPE} = \frac{100}{|\Omega|} \sum_{i \in \Omega} \left| \frac{\hat{Y}_i - Y_i}{\max(|Y_i|, \epsilon)} \right|, \tag{39}$$

where $|\Omega|$ is the number of valid target positions and $\epsilon = 10^{-5}$ is used for numerical stability in MAPE. Following common practice in traffic forecasting, MAE is used as the primary metric, while RMSE and MAPE are reported as complementary metrics. Each reported result is the mean and sample standard deviation over three random seeds: 2024, 2025, and 2026.

5.4 Standard protocol without additional synthetic masking

We first evaluate all methods without injecting additional synthetic missingness. Dataset-native null readings remain in the supplied data, and ACSC-GCRN receives an all-one historical observation-availability mask under this standard protocol. Table 2 therefore evaluates standard forecasting performance and provides a reference for the subsequent missing-input experiments.

ACSC-GCRN achieves the best results on all three metrics for PEMS04. For PEMS08, it ranks first in MAE and MAPE and second in RMSE. On METR-LA, HD-TTS obtains the best results, while ACSC-GCRN ranks second on all three metrics. These results show that ACSC-GCRN remains competitive with both general forecasting models and methods designed for incomplete traffic data when no additional synthetic masking is applied.

Table 2 Forecasting performance under the standard protocol without additional synthetic masking. Results are reported as mean ± sample standard deviation over three runs. The best result is shown in bold and the second-best result is underlined

Dataset	Model	MAE	RMSE	MAPE (%)
METR-LA	AGCRN	3.3187 ± 0.0121	6.8002 ± 0.0200	9.3466 ± 0.0981
	DLinear	4.8208 ± 0.0018	10.1536 ± 0.0113	12.5061 ± 0.0108
	NLinear	4.9635 ± 0.0015	10.8837 ± 0.0031	12.1159 ± 0.0079
	STID	3.6426 ± 0.0004	7.5334 ± 0.0095	10.4131 ± 0.0889
	StemGNN	3.8225 ± 0.3045	7.4657 ± 0.4133	10.7466 ± 0.7290
	TimesNet	4.4205 ± 0.0060	9.5910 ± 0.0056	11.5263 ± 0.0533
	PatchTST	4.3027 ± 0.0058	9.6029 ± 0.0132	11.3424 ± 0.0457
	iTransformer	4.2033 ± 0.0048	9.3950 ± 0.0202	11.0610 ± 0.0261
	BiTGraph	3.2265 ± 0.0109	6.5122 ± 0.0362	9.0590 ± 0.0685
	HD-TTS	**3.1140 ± 0.0119**	**6.2315 ± 0.0445**	**8.6486 ± 0.0350**
	ACSC-GCRN	3.1628 ± 0.0058	6.3532 ± 0.0525	8.6979 ± 0.1534
PEMS04	AGCRN	20.3580 ± 0.1194	32.2441 ± 0.1961	13.9571 ± 0.2169
	DLinear	27.9870 ± 0.0010	43.8979 ± 0.0140	19.1761 ± 0.0471
	NLinear	28.2886 ± 0.0006	44.4581 ± 0.0087	18.6321 ± 0.0075
	STID	23.5449 ± 0.0349	36.7473 ± 0.0577	16.9339 ± 0.5410
	StemGNN	23.0260 ± 1.0200	35.4479 ± 1.1763	17.1887 ± 1.4083
	TimesNet	21.0707 ± 0.0554	33.9763 ± 0.0493	14.1818 ± 0.0649
	PatchTST	26.8357 ± 0.0234	41.9902 ± 0.0323	17.8961 ± 0.0266
	iTransformer	24.0503 ± 0.0107	38.6069 ± 0.0034	15.9141 ± 0.0307
	BiTGraph	20.5063 ± 0.0595	32.3607 ± 0.0557	14.2721 ± 0.2493
	HD-TTS	20.9623 ± 0.0163	32.7507 ± 0.0197	14.5561 ± 0.1024
	ACSC-GCRN	**19.5809 ± 0.0844**	**31.4174 ± 0.1223**	**13.3382 ± 0.1500**
PEMS08	AGCRN	16.6620 ± 0.0932	26.0929 ± 0.2981	10.9585 ± 0.1989
	DLinear	22.4276 ± 0.0024	35.4376 ± 0.0424	14.6338 ± 0.2066
	NLinear	22.6142 ± 0.0031	35.8697 ± 0.0148	13.6780 ± 0.0076
	STID	18.7555 ± 0.0391	29.2139 ± 0.0711	12.0538 ± 0.2356
	StemGNN	17.7703 ± 0.1187	27.4243 ± 0.3214	12.4101 ± 0.3744
	TimesNet	17.2981 ± 0.1181	27.6775 ± 0.1584	10.7870 ± 0.0526
	PatchTST	20.6285 ± 0.0263	33.2529 ± 0.0776	12.7353 ± 0.0412
	iTransformer	18.3837 ± 0.0880	29.9157 ± 0.1413	11.5110 ± 0.0412
	BiTGraph	16.6277 ± 0.1117	25.9433 ± 0.2387	10.9062 ± 0.2216
	HD-TTS	16.3496 ± 0.0634	**25.4353 ± 0.1053**	10.7416 ± 0.1290
	ACSC-GCRN	**16.1313 ± 0.1056**	25.4877 ± 0.0991	**10.3712 ± 0.1152**

Relative to AGCRN, which uses the same graph-convolutional recurrent backbone, ACSC-GCRN reduces MAE by 4.70%, 3.82%, and 3.19% on METR-LA, PEMS04, and PEMS08, respectively. This comparison indicates that the proposed availability-aware input encoding and multi-source spatial modeling do not trade away standard-protocol forecasting accuracy. The result provides a reference for evaluating robustness under the synthetic and dataset-native missing-input protocols considered below.

5.5 Robustness under synthetic missingness

We next evaluate forecasting robustness under random, block, and node-level missingness at rates from 10% to 50%. For each model, dataset, missingness pattern, and metric, we first average the results of the three random seeds at each missing rate and then take an equal-weight macro-average over the five rates. Tables 3 compares ACSC-GCRN with the baseline attaining the lowest macro-averaged error among the ten competing methods.

Table 3 Macro-averaged forecasting performance under synthetic missingness. The relative gain is calculated as $(E_{\text{base}} - E_{\text{ACSC}})/E_{\text{base}} \times 100\%$; a negative value indicates that the selected baseline performs better

Dataset	Pattern	Metric	Best baseline	Baseline	ACSC-GCRN	Gain
METR-LA	Random	MAE	HD-TTS	**3.1734**	3.2005	−0.85%
		RMSE	HD-TTS	**6.3182**	6.4652	−2.33%
		MAPE (%)	HD-TTS	8.8521	**8.8125**	+0.45%
	Block	MAE	HD-TTS	**3.1783**	3.2018	−0.74%
		RMSE	HD-TTS	**6.3455**	6.4526	−1.69%
		MAPE (%)	HD-TTS	8.8648	**8.7729**	+1.04%
	Node	MAE	HD-TTS	**3.2552**	3.2655	−0.32%
		RMSE	HD-TTS	**6.4513**	6.5851	−2.07%
		MAPE (%)	HD-TTS	9.1145	**8.9396**	+1.92%
PEMS04	Random	MAE	AGCRN	21.0610	**19.9560**	+5.25%
		RMSE	AGCRN	33.1199	**32.0120**	+3.35%
		MAPE (%)	AGCRN	14.7237	**13.5878**	+7.71%
	Block	MAE	AGCRN	21.0886	**19.9539**	+5.38%
		RMSE	AGCRN	33.1748	**31.9278**	+3.76%
		MAPE (%)	TimesNet	14.1848	**13.6982**	+3.43%
	Node	MAE	AGCRN	22.5414	**20.8786**	+7.38%
		RMSE	BiTGraph	35.4511	**33.5752**	+5.29%
		MAPE (%)	BiTGraph	15.9795	**14.3503**	+10.20%
PEMS08	Random	MAE	HD-TTS	16.9842	**16.5514**	+2.55%
		RMSE	HD-TTS	26.2251	**26.0454**	+0.69%
		MAPE (%)	HD-TTS	11.2102	**10.6801**	+4.73%
	Block	MAE	HD-TTS	17.1789	**16.6833**	+2.88%
		RMSE	HD-TTS	26.5437	**26.1848**	+1.35%
		MAPE (%)	TimesNet	11.0884	**10.7394**	+3.15%
	Node	MAE	HD-TTS	19.8492	**17.9859**	+9.39%
		RMSE	HD-TTS	32.0065	**29.3121**	+8.42%
		MAPE (%)	HD-TTS	12.6516	**11.6457**	+7.95%

ACSC-GCRN outperforms the strongest baseline in 21 of the 27 dataset–pattern–metric aggregates. It achieves the lowest aggregate error in all nine comparisons on PEMS04 and all nine comparisons on PEMS08. On METR-LA, HD-TTS obtains lower MAE and RMSE under all three missingness patterns, whereas ACSC-GCRN obtains lower MAPE. The largest relative gains are observed under node-level missingness, including a 10.20% MAPE reduction on PEMS04 and a 9.39% MAE reduction on PEMS08. These results indicate that the proposed spatial compensation mechanism is particularly beneficial on PEMS04 and PEMS08 when a node loses its complete local history.

Considering all 135 individual synthetic-missingness comparisons before macro-averaging (three datasets, three patterns, five missing rates, and three metrics), ACSC-GCRN ranks first in 102 cases and second in the remaining 33 cases. Together with the nine standard-protocol comparisons in Table 2, ACSC-GCRN is the only evaluated method that remains within the top two in all 144 comparisons. Therefore, the results support the cross-condition robustness of ACSC-GCRN while also showing that it does not uniformly outperform HD-TTS on METR-LA.

5.6 Sensitivity to missing rate

To examine the effect of missingness severity, we aggregate the results at each missing rate separately. For every dataset, rate, model, and metric, we first average over the three missingness patterns and three random seeds. We then select the baseline with the

lowest aggregated error from the ten competing methods and calculate the relative gain of ACSC-GCRN against that baseline. The baseline selection is performed after aggregation, so each reported cell compares ACSC-GCRN with one consistent baseline across the three patterns. The resulting relative gains are reported in Table 4.

On PEMS04, the gains of ACSC-GCRN are positive and increase monotonically with the missing rate for all three metrics. On PEMS08, the MAE and RMSE gains also increase monotonically, while the MAPE gains remain positive but fluctuate across rates. These results show that the relative advantage of ACSC-GCRN generally becomes more pronounced as the historical input becomes less complete on the two flow datasets.

On METR-LA, ACSC-GCRN initially trails the strongest baseline, but the gap narrows as the missing rate increases. Its MAE comparison changes from −1.37% at a 10% missing rate to a 0.14% gain at 50%, while its RMSE gap decreases from 2.37% to 1.71%. Its MAPE becomes better than the strongest baseline from the 30% missing rate onward. Overall, the results support increasing relative robustness under more severe missingness, but they do not imply a monotonic improvement for every dataset and metric.

5.7 Evaluation under the dataset-native invalid-reading protocol

Controlled synthetic masking cannot represent every observation condition present in traffic datasets. We therefore conduct an additional experiment on METR-LA using the zero-coded invalid readings identified by the dataset null-value convention $v_{\text{null}} = 0.0$, without introducing any synthetic missingness. These zero-coded positions account for approximately 8.11% of the raw METR-LA data, consistent with the proportion reported in [17]. Under this protocol, they are treated as unavailable according to the dataset convention; however, the convention itself does not identify the physical cause of each zero-coded reading.

For ACSC-GCRN, BiTGraph, and HD-TTS, the historical availability mask is derived directly from the raw METR-LA inputs according to this convention. The numerical inputs and the established normalization procedure remain unchanged, and no additional synthetic mask is applied. General forecasting baselines do not consume a historical mask; because the native protocol does not change their numerical inputs and invalid future targets are already excluded from loss and metric computation, their standard-protocol METR-LA results are reused. All reported methods are therefore evaluated on the same valid future target positions.

As shown in Table 5, ACSC-GCRN ranks second on MAE and RMSE and first on MAPE. Its MAE is 0.68% higher than that of HD-TTS and 2.16% lower than that of

Table 4 Relative gain (%) of ACSC-GCRN over the strongest baseline at each synthetic missing rate. Results are aggregated over three missingness patterns and three random seeds

Dataset	Metric	10%	20%	30%	40%	50%
METR-LA	MAE	−1.37	−1.02	−0.64	−0.32	+0.14
	RMSE	−2.37	−2.07	−2.01	−2.00	−1.71
	MAPE	−0.14	−0.05	+1.55	+2.15	+2.11
PEMS04	MAE	+4.65	+5.15	+5.98	+6.64	+7.59
	RMSE	+3.59	+4.34	+5.19	+5.26	+5.70
	MAPE	+7.60	+7.73	+9.94	+10.05	+11.26
PEMS08	MAE	+3.21	+4.23	+4.98	+5.93	+7.20
	RMSE	+1.77	+3.02	+3.77	+4.50	+5.71
	MAPE	+5.67	+4.69	+8.46	+4.05	+8.03

Table 5 Forecasting performance under the METR-LA dataset-native invalid-reading protocol. Results are mean ± sample standard deviation over three random seeds. The best result is shown in bold and the second-best result is underlined

Model	MAE	RMSE	MAPE (%)
HD-TTS	**3.1147 ± 0.0024**	**6.2147 ± 0.0191**	<u>8.6680 ± 0.0802</u>
ACSC-GCRN	<u>3.1360 ± 0.0075</u>	<u>6.3176 ± 0.0542</u>	**8.6109 ± 0.0579**
BiTGraph	3.2051 ± 0.0092	6.4917 ± 0.0392	9.0280 ± 0.1109
AGCRN	3.3187 ± 0.0121	6.8002 ± 0.0200	9.3466 ± 0.0981
STID	3.6426 ± 0.0004	7.5334 ± 0.0095	10.4131 ± 0.0889
StemGNN	3.8225 ± 0.3045	7.4657 ± 0.4133	10.7466 ± 0.7290
iTransformer	4.2033 ± 0.0048	9.3950 ± 0.0202	11.0610 ± 0.0261
PatchTST	4.3027 ± 0.0058	9.6029 ± 0.0132	11.3424 ± 0.0457
TimesNet	4.4205 ± 0.0060	9.5910 ± 0.0056	11.5263 ± 0.0533
DLinear	4.8208 ± 0.0018	10.1536 ± 0.0113	12.5061 ± 0.0108
NLinear	4.9635 ± 0.0015	10.8837 ± 0.0031	12.1159 ± 0.0079

BiTGraph. Thus, ACSC-GCRN remains competitive when observation availability is obtained from dataset-native invalid readings rather than a controlled synthetic mask, while the results do not imply that it is uniformly superior on every metric.

5.8 Ablation study

We examine the mechanisms directly associated with the proposed availability-conditioned forecasting framework. The full model is compared with six targeted variants on PEMS04 and PEMS08 under random and node-level missingness at rates 0.3 and 0.5. Each setting is repeated with three random seeds, resulting in 168 runs for the core ablation analysis. The variants test the explicit availability mask, gated versus additive input encoding, availability-conditioned versus global or equal-weight graph fusion, and the contributions of the correlation and adaptive graphs.

The Full variant employs the gated availability-aware encoder and availability-conditioned dynamic fusion of all three graph sources. The w/o mask variant removes the explicit availability signal, while additive encoding replaces the gated encoder with a direct addition of the projected value and mask representations. Global fusion uses one set of learnable graph-source weights shared across the dataset, whereas equal-weight fusion assigns the same weight to every graph source. The w/o correlation and w/o adaptive variants remove the corresponding graph source while retaining the remaining model structure. All other architectural and training settings are kept unchanged.

Table 6 reports the complete metrics under random missingness at rate 0.5 as a challenging missing-input setting. The paired ΔMAE is computed against the full model using the same dataset, missingness condition, and random seed; a positive value indicates that the variant performs worse. The conclusions are further verified over all eight dataset–pattern–rate settings in Table 7.

To avoid drawing conclusions from one representative condition, Table 7 summarizes the paired MAE changes over all eight dataset–pattern–rate settings. Each pattern–rate entry is the equal-weight average over PEMS04 and PEMS08, while the overall column is the average over all eight settings.

Removing the explicit availability mask increases MAE by 0.95% on average and degrades all 8/8 settings. Replacing the gated encoder with additive encoding increases MAE by 0.83% and also degrades all settings. These results support the use of availability

Table 6 Ablation results under random missingness at rate 0.5. Results are mean ± sample standard deviation over three seeds. Positive ΔMAE indicates degradation relative to the full model

Dataset	Variant	MAE	RMSE	MAPE (%)	ΔMAE
PEMS04	Full	20.2842 ± 0.0873	32.5412 ± 0.3171	13.8117 ± 0.1486	0.00%
	w/o mask	20.3568 ± 0.0031	32.6490 ± 0.1459	13.9399 ± 0.1185	+0.36%
	additive encoding	20.3950 ± 0.0187	32.8876 ± 0.0959	13.9076 ± 0.1835	+0.55%
	global fusion	20.3700 ± 0.0240	32.6974 ± 0.1237	13.8301 ± 0.0475	+0.42%
	equal-weight fusion	20.4890 ± 0.0397	32.6521 ± 0.1598	14.0784 ± 0.1345	+1.01%
	w/o correlation	20.6460 ± 0.0332	32.7131 ± 0.1868	14.2922 ± 0.2526	+1.79%
	w/o adaptive	20.3804 ± 0.0641	32.7794 ± 0.2804	13.8522 ± 0.1644	+0.48%
PEMS08	Full	16.9240 ± 0.1079	26.6307 ± 0.1310	10.8912 ± 0.1369	0.00%
	w/o mask	17.2172 ± 0.0706	26.9800 ± 0.1181	11.3564 ± 0.3841	+1.73%
	additive encoding	17.0977 ± 0.0541	26.8100 ± 0.1631	10.9552 ± 0.2009	+1.03%
	global fusion	16.9401 ± 0.1419	26.6159 ± 0.2514	10.9178 ± 0.1760	+0.10%
	equal-weight fusion	17.0873 ± 0.2007	26.7143 ± 0.3151	10.9715 ± 0.0633	+0.97%
	w/o correlation	17.2789 ± 0.0450	26.9833 ± 0.0573	11.2112 ± 0.2802	+2.10%
	w/o adaptive	17.0646 ± 0.0393	26.7970 ± 0.0625	10.9148 ± 0.1236	+0.83%

Table 7 Cross-condition paired ΔMAE (%) of the core ablation variants. R and N denote random and node-level missingness, respectively. Positive values indicate degradation; "worse" counts the settings in which the variant has higher mean MAE than the full model

Variant	R–0.3	R–0.5	*N*–0.3	*N*–0.5	Overall	Worse
w/o mask	+0.684%	+1.046%	+0.994%	+1.087%	+0.953%	8/8
additive encoding	+0.391%	+0.788%	+1.128%	+1.022%	+0.832%	8/8
global fusion	+0.002%	+0.261%	+0.172%	+0.898%	+0.333%	7/8
equal-weight fusion	+0.493%	+0.991%	+1.190%	+1.842%	+1.129%	7/8
w/o correlation	+1.282%	+1.943%	+3.622%	+4.979%	+2.957%	8/8
w/o adaptive	+0.380%	+0.655%	+0.822%	+0.496%	+0.588%	8/8

as an explicit input signal and show that its interaction with the value representation is better modeled by the learned gate than by a fixed addition.

Replacing availability-conditioned fusion with one global graph mixture produces a smaller average MAE increase of 0.33% and degrades 7/8 settings. Its mean degradation increases from 0.09% at missing rate 0.3 to 0.58% at rate 0.5, indicating that the benefit of conditional fusion is limited but becomes more evident under more severe missingness. Equal-weight fusion increases MAE by 1.13% on average and degrades 7/8 settings, supporting learned graph-source allocation over uniform averaging.

Removing the correlation graph causes the largest stable degradation among the graph-source ablations, increasing MAE by 2.96% on average across all 8/8 settings. Removing the adaptive graph increases MAE by 0.59% and also degrades all settings. Therefore, the correlation graph provides the main data-driven compensation, whereas the adaptive graph offers a smaller but consistent complementary contribution. The results demonstrate stable aggregate effects without implying that the full model must be best on every individual metric or condition.

5.9 Implementation details

All experiments use 12 historical time steps to predict the next 12 time steps. For a controlled comparison, all models, including ACSC-GCRN, are trained for 50 epochs with a batch size of 64 using the Adam optimizer. The learning rate is uniformly set to 0.001, the weight decay is set to zero, and the gradient norm is clipped at 5.0. The checkpoint with the lowest validation MAE is used for test-set evaluation. Each experimental condition is repeated with random seeds 2024, 2025, and 2026, and the results are reported as the mean and sample standard deviation over the three runs.

For ACSC-GCRN, the hidden-state dimension is 64, the recurrent encoder contains two layers, the Chebyshev polynomial order is 2, and the node-embedding dimension is 10. The correlation graph retains the 20 largest absolute Pearson correlations for each node, computed exclusively from the training split. The maximum temporal block length used for block missingness is six time steps. Normalization statistics are also estimated only from the training split, and all compared methods use the same deterministic synthetic mask for a given dataset, sample, missingness condition, and seed.

AGCRN is implemented using the standard adaptive graph convolutional recurrent formulation, while the remaining general forecasting baselines use the corresponding BasicTS model components. BiTGraph and HD-TTS are adapted from their official open-source implementations to the same data splits, input–output horizons, missingness protocol, metric definitions, and training budget. Because BiTGraph's original temporal kernels are designed for a 24-step input, its kernel set is adjusted to $(2, 3, 5, 6)$ to

obtain a valid receptive field for the 12-step history. Experiments are conducted on a Linux workstation equipped with NVIDIA GeForce RTX 4090D GPUs. The software environment includes Python 3.11.15, PyTorch 2.5.1 with CUDA 12.4, and BasicTS 1.1.0.

6 Discussion

6.1 Availability-conditioned feature Use and spatial compensation

The experimental findings support the central view of this work that observation availability should influence both feature use and spatial compensation. At the feature level, the availability mask prevents placeholder values at unavailable positions from contributing to the value branch, while the learned gate regulates how available observations are used and the availability representation is retained. The consistent degradation caused by removing the mask or replacing the gated encoder with additive encoding demonstrates that these roles cannot be reproduced by placeholder values or a fixed value–mask combination alone. Importantly, the learned gate is a feature-control mechanism rather than a physically calibrated confidence score; estimating the measurement uncertainty of available observations is outside the scope of this study.

At the spatial level, physical, statistical, and latent relations provide complementary cross-node information when a node's own historical observations are incomplete. The ablation results show that the correlation graph provides the most prominent data-driven compensation, while the adaptive graph offers a smaller but stable complementary contribution. Availability-conditioned fusion provides an additional improvement over global fusion, with its aggregate advantage becoming more pronounced at higher missing rates. This result highlights the value of adapting spatial compensation to observation availability under more challenging incomplete-history conditions, without implying that all graph sources contribute equally in every setting.

6.2 Cross-condition robustness and dataset-specific behavior

Across the standard and synthetic-missingness protocols, ACSC-GCRN is the only evaluated method that remains within the top two in all 144 dataset–condition–metric comparisons, ranking first in 102 of the 135 synthetic-missingness comparisons. Its advantages are particularly evident on PEMS04 and PEMS08, under node-level missingness, and at higher missing rates, where spatial compensation becomes more important as nodes lose access to their own historical observations. At the same time, its competitive results under the standard protocol show that the robustness to incomplete histories is not obtained at the expense of ordinary forecasting performance. Thus, the main empirical strength of ACSC-GCRN is its stable performance across diverse observation conditions rather than universal superiority in every individual setting.

HD-TTS performs better on METR-LA under the standard protocol and retains lower MAE and RMSE under synthetic missingness. Because its advantage is already present without additional synthetic masking, HD-TTS may be particularly well suited to the overall characteristics of METR-LA rather than benefiting only from its treatment of incomplete observations. Nevertheless, the gap between ACSC-GCRN and the strongest baseline generally narrows as the missing rate increases, and ACSC-GCRN achieves lower aggregated MAPE on METR-LA under synthetic missingness. These results reveal dataset-specific strengths while supporting the broader cross-condition robustness of ACSC-GCRN.

The METR-LA dataset-native invalid-reading experiment further complements the controlled synthetic tests. Without introducing additional synthetic missingness, ACSC-GCRN ranks second on MAE and RMSE and first on MAPE when availability is derived from the dataset's zero-coded invalid-reading convention. This result shows that the method remains competitive beyond the controlled missingness patterns. However, the convention does not identify the physical cause of each invalid reading, and evidence from one dataset should not be interpreted as a universal claim about all forms of real-world missingness.

6.3 Limitations and future work

Although the experiments cover multiple datasets, missingness patterns, and missing rates, real traffic systems may exhibit more diverse observation conditions than those represented by controlled synthetic masks and one dataset-native invalid-reading convention. Future work will evaluate the method on broader traffic networks with explicit observation-status annotations and investigate finer-grained availability conditioning for richer spatiotemporal observation patterns. These extensions will further examine the generalizability of availability-conditioned spatial compensation in practical forecasting scenarios.

7 Conclusion

This paper presented ACSC-GCRN for direct traffic forecasting from incomplete historical observations. The method treats missingness as an observation-unavailability state rather than an ordinary numerical value. An availability-gated encoder prevents placeholder values at unavailable positions from contaminating the value representation while retaining the availability information. Based on this representation, a multi-source spatial compensation module integrates physical, statistical, and latent spatial relations through sample–node-specific weights conditioned on historical observation availability and node embeddings. The resulting spatial support is used by a graph convolutional recurrent backbone to produce future traffic states without a separate history-imputation stage.

Experiments across three datasets, multiple missingness patterns and rates, general forecasting baselines, and methods designed for incomplete traffic data demonstrate the stable cross-condition performance of ACSC-GCRN. The model remains consistently competitive under both the standard and synthetic-missingness protocols and also performs well under the METR-LA dataset-native invalid-reading convention. Ablation results confirm the value of explicit availability representation and gated input encoding. They also show that the correlation graph provides the most prominent data-driven compensation, the adaptive graph offers a smaller but stable complement, and availability-conditioned fusion provides a more evident advantage over global fusion as missingness becomes more severe. Overall, the results support using observation availability not only to control historical feature use but also to guide the selection of spatial compensation sources for robust traffic forecasting with incomplete histories.

Author contributions

Zhouyi Xu contributed to the conceptualization and methodology of the study, model and software development, data preparation, experimental implementation, formal analysis, validation, visualization, and preparation of the original manuscript; Wujian Yang contributed to methodology, experimental validation, formal analysis, investigation, and critical review and revision of the manuscript; Guanlin Chen contributed to conceptualization, research supervision, project

administration, provision of resources, validation of the research design and experimental results, and critical review and revision of the manuscript.

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Data availability
The datasets analyzed in this study are publicly available. Publicly accessible copies of the benchmark datasets can be obtained from the following repositories: METR-LA is available at https://doi.org/10.5281/zenodo.5146275, while PEMS04 and PEMS08 are available at https://doi.org/10.5281/zenodo.7816008.

Declarations

Ethical approval
Not applicable.

Consent to participate
Not applicable.

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Not applicable.

Competing interests
The authors declare no competing interests.

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